

Learning target AD3, version 1

A wizard is developing a shrinking spell, but he needs to make sure it will work with his pointy hat, which is a cone whose height is three times its radius. Once he casts the spell, the height of his hat shrinks by 2 inches every second. How fast is the volume of his hat changing when the hat is just 3 inches tall? Give units on your answer.

Hints: Draw three pictures, label variables and constants, write down what you know and what you want to know, relate the variables, find the derivative wrt time, substitute and solve.

The volume of a cone is a third of the volume of the cylinder that contains it: $V = 1/3 \cdot \pi r^2 h$.

Learning target AD3, version 2

Gravel is being dumped from a conveyor belt at a rate of 12 cubic feet per minute. The gravel forms a pile in the shape of a right circular cone whose base diameter and height are always the same. How fast is the height of the pile increasing when the pile is 12 ft high?

Learning targets DF1 and DFa, version 2

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table¹ gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t .

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.
Write a sentence explaining what the number means about how the area of the glacier is changing.
Don't say "per," and don't say "rate."
2. Compute an approximation for $A'(1999)$, and **include units** for this number.
Do you think your approximation is too high or too low? Why?
3. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

¹Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research*, 42(2), 198-209.

Learning target DF2, version 1

Suppose that $f(x) = 3x^2 - 5x + 4$.

1. Use the limit definition of the derivative to find $f'(x)$. **No shortcut rules!**

2. Evaluate at $x = 8$.

3. (Bonus!) What happens to the 3? What about the -5 ? And the 4?

Learning target DF2, version 2

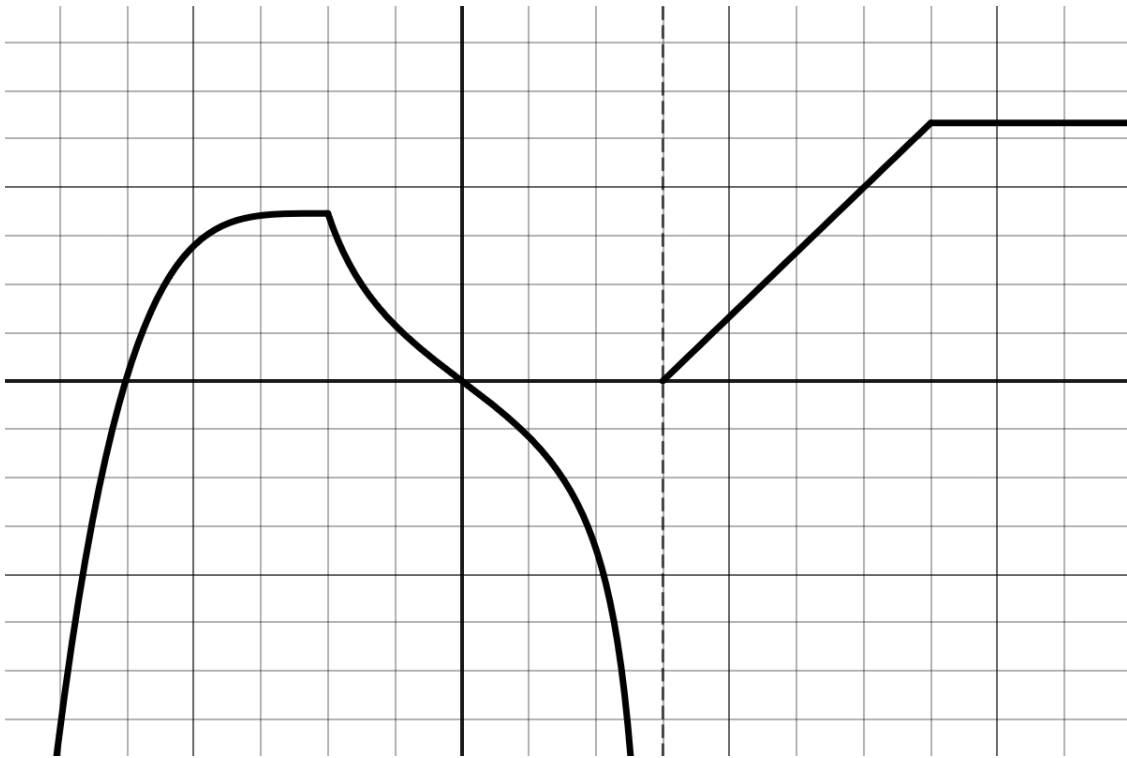
Suppose that $g(w) = 6w^3 - 2w^2 - 9w - 2$. Use the limit definition of the derivative to find $g'(w)$.

Algebra hint: $(w + h)^3 = w^3 + 3w^2h + 3wh^2 + h^3$.

PS - The answer is $g'(w) = 18w^2 - 4w + 9$.

Learning target DFb, version 2

Here is the graph of some wacky function $h(t)$:



Sketch the graph of $h'(t)$ on the blank axes below.

